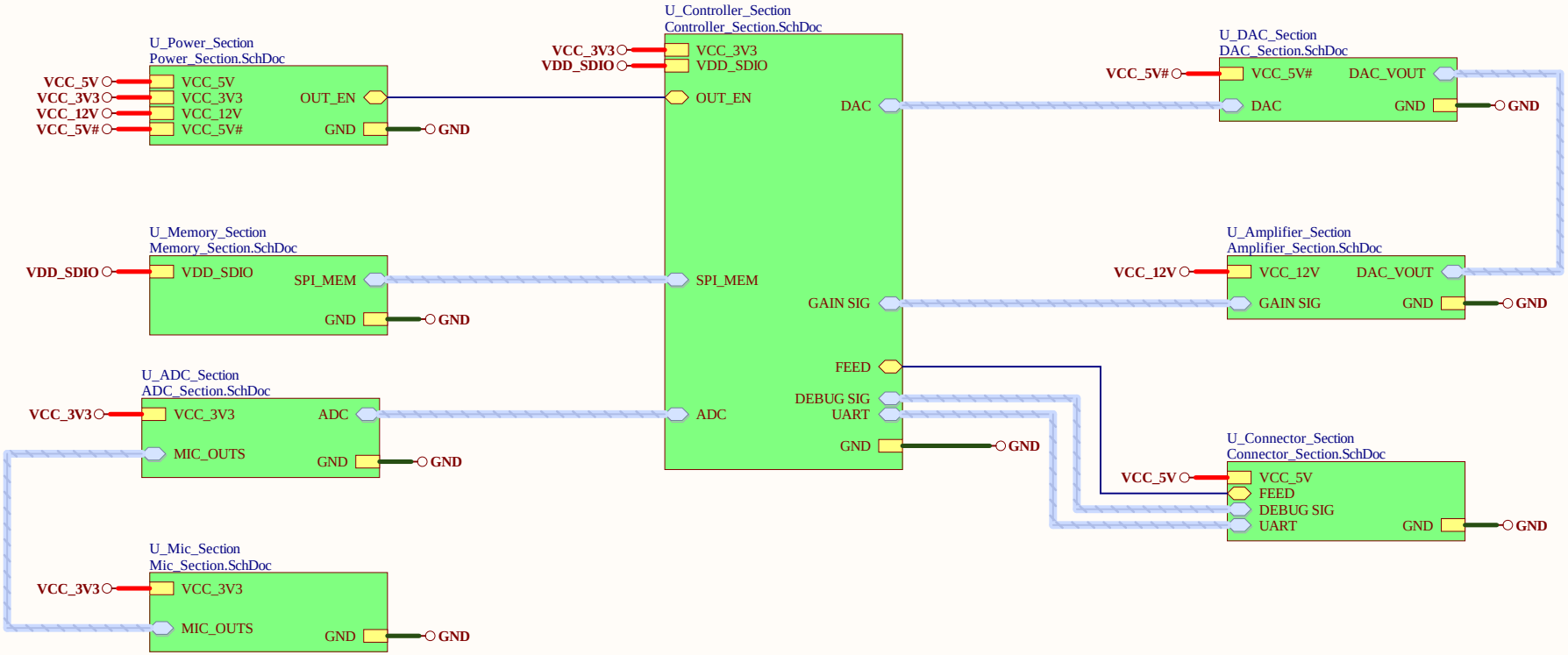
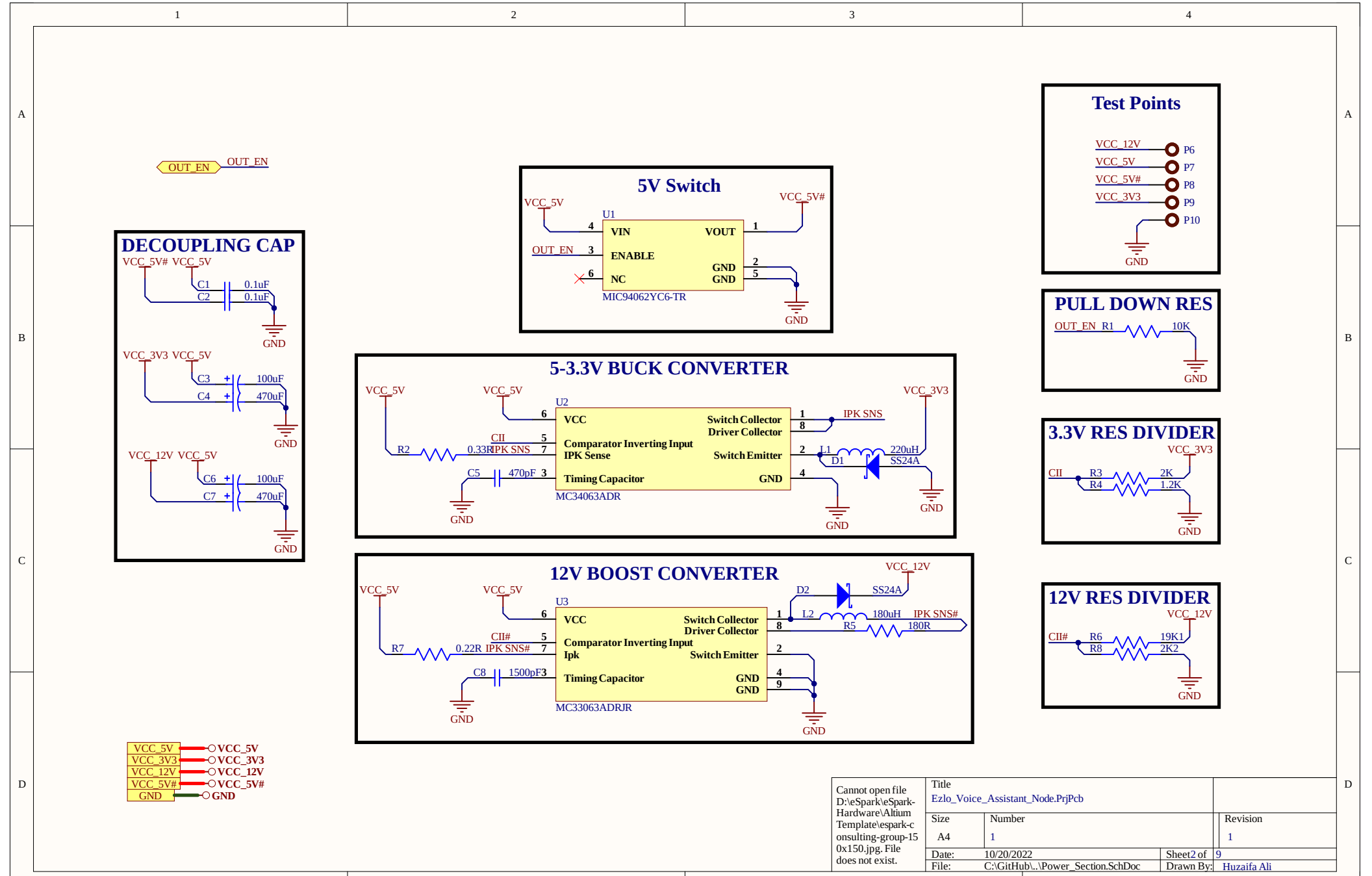
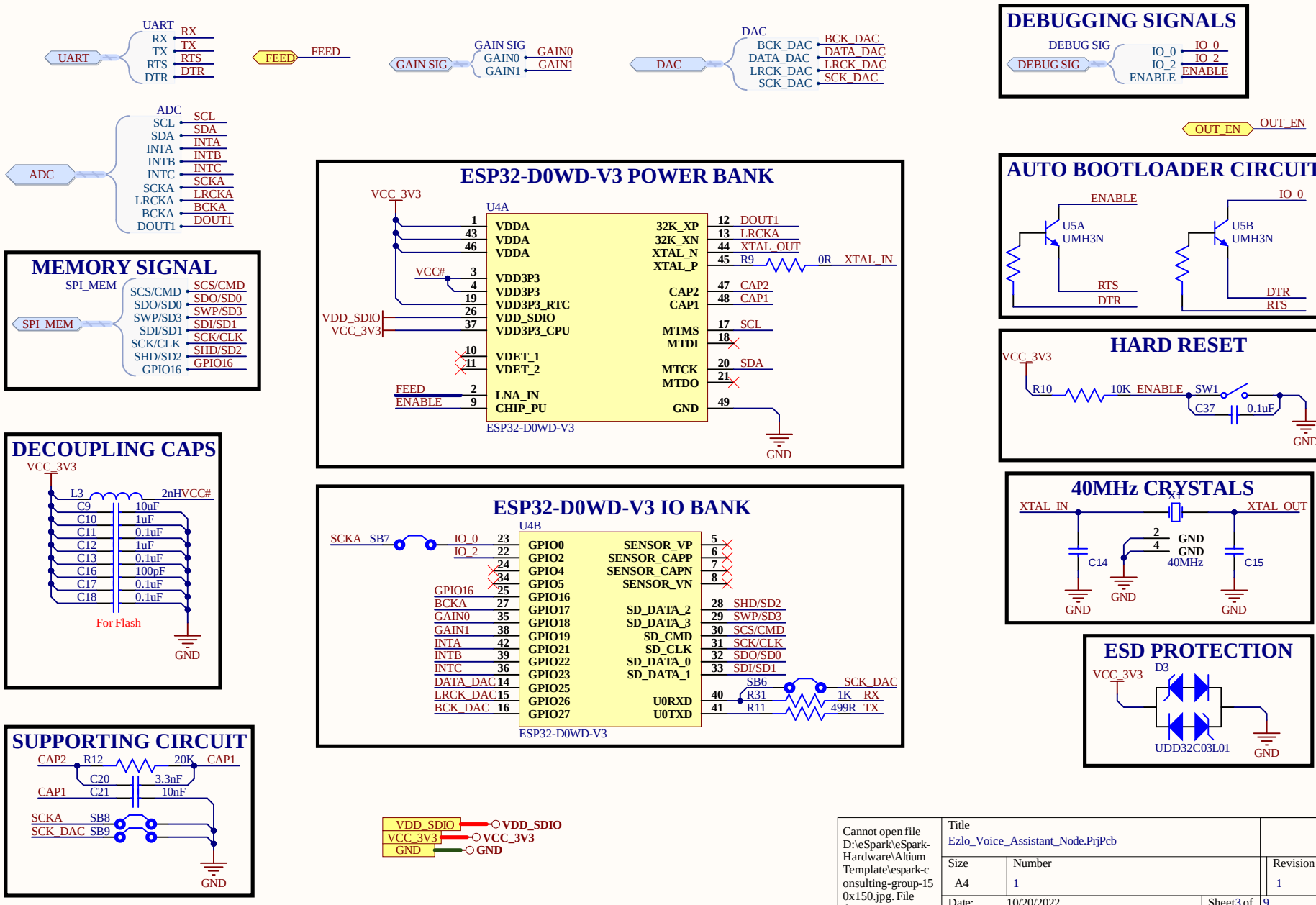


Cannot open file D:\eSpark\eSpark-Hardware\Altium Template\espark-consulting-group-150x150.jpg. File does not exist.

Title	Ezlo_Voice_Assistant_Node.PrjPcb		
Size	Number		Revision
A4	1		1
Date:	10/20/2022	Sheet 1 of	9
File:	C:\GitHub\..\Top_Sheet.SchDoc	Drawn By:	Huzaifa Ali

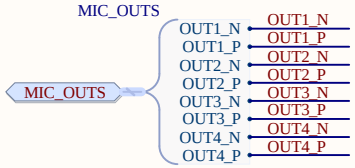


Cannot open file D:\eSpark\eSpark-Hardware\Altium Template\espark-consulting-group-150x150.jpg. File does not exist.	Title: Ezlo_Voice_Assistant_Node.PrjPcb		
	Size: A4	Number: 1	Revision: 1
	Date: 10/20/2022	Sheet 2 of 9	
	File: C:\GitHub\..\Power_Section.SchDoc	Drawn By: Huzaifa Ali	

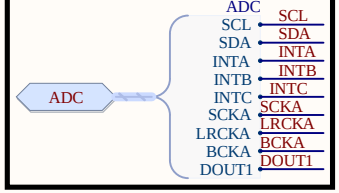


Cannot open file D:\eSpark\eSpark-Hardware\Altium Template\espark-consulting-group-150x150.jpg. File does not exist.	Title: Ezlo_Voice_Assistant_Node.PrjPcb	
	Size: A4 / Number: 1	Revision: 1
	Date: 10/20/2022	Sheet 3 of 9
	File: C:\GitHub\..\Controller_Section.SchDoc	Drawn By: Huzaifa Ali

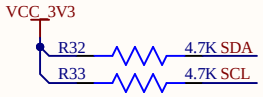
MIC OUT SIGNALS



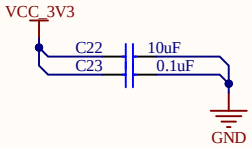
ADC SIGNALS



I2C Pull-up

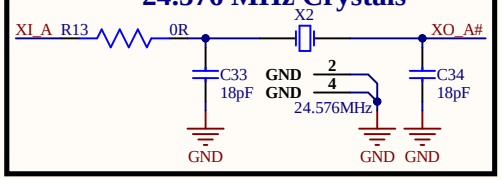


DECOUPLING CAPS

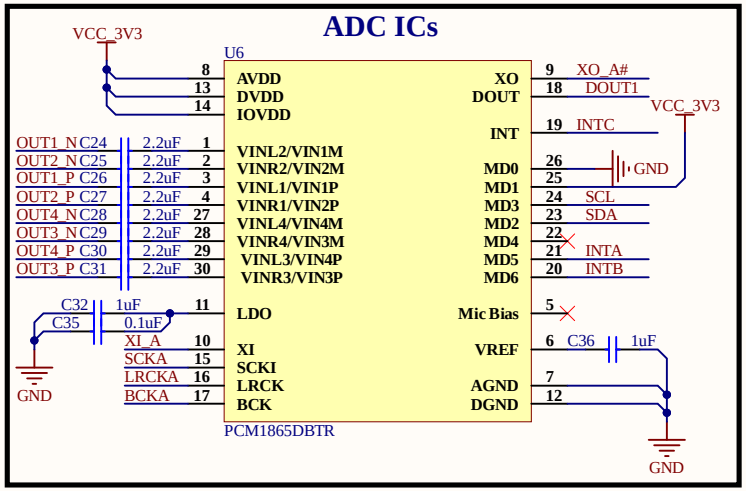


A minimum value of 2.2 µF is recommended in Figure 7 for codecs, which may have a very low input impedance at some PGA gain settings.
Datasheet page: 10

24.576 MHz Crystals



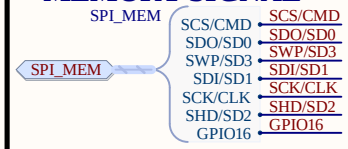
ADC ICs



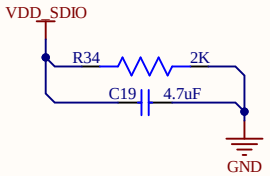


Cannot open file D:\eSpark\eSpark-Hardware\Altium Template\espark-consulting-group-150x150.jpg. File does not exist.	Title: Ezlo_Voice_Assistant_Node.PrjPcb		
	Size: A4	Number: 1	Revision: 1
	Date: 10/20/2022	Sheet 4 of 9	
	File: C:\GitHub\..\ADC_Section.SchDoc	Drawn By: Huzaifa Ali	

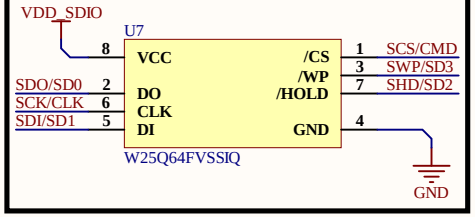
MEMORY SIGNAL



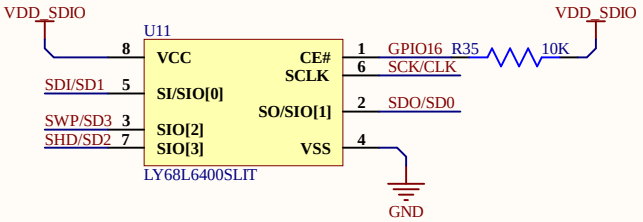
SUPPORTING CIRCUIT



64 Mbit SPI FLASH

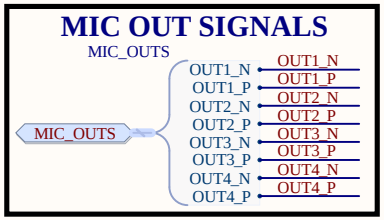


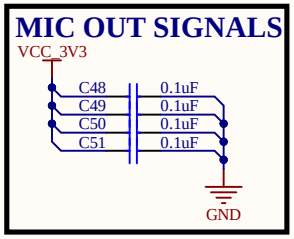
64 Mbit PSRAM



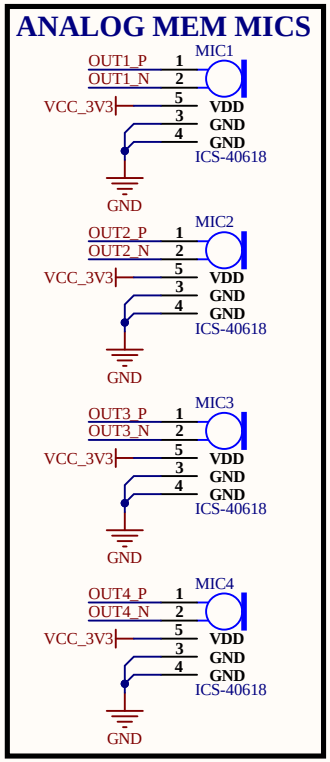


Cannot open file D:\eSpark\eSpark-Hardware\Altium Template\espark-consulting-group-150x150.jpg. File does not exist.	Title: Ezlo_Voice_Assistant_Node.PrjPcb		
	Size: A4	Number: 1	Revision: 1
	Date: 10/20/2022	Sheet 5 of 9	
	File: C:\GitHub\..\Memory_Section.SchDoc	Drawn By: Huzaifa Ali	









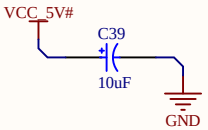
Cannot open file D:\eSpark\eSpark-Hardware\Altium Template\espark-consulting-group-150x150.jpg. File does not exist.	Title: Ezlo_Voice_Assistant_Node.PrjPcb		
	Size: A4	Number: 1	Revision: 1
	Date: 10/20/2022	Sheet 6 of 9	
	File: C:\GitHub\..\Mic_Section.SchDoc	Drawn By: Huzaifa Ali	



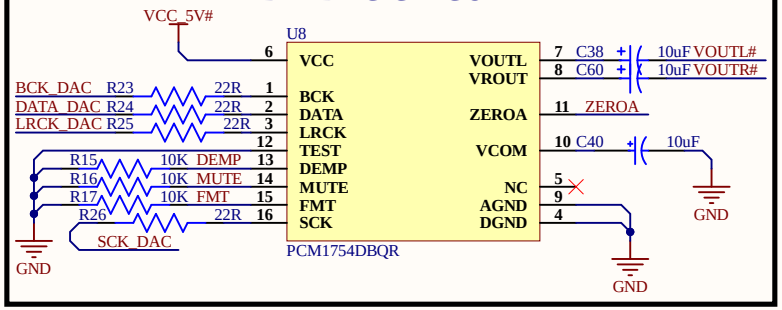


Input impedance of amplifier is 15K
when GAIN0-=0 and GAIN1=1
fC = 1/(2π × C × R)
Cutt of frequency of low pass filter is 10Hz
Range of sound signals= 20Hz-20KHz

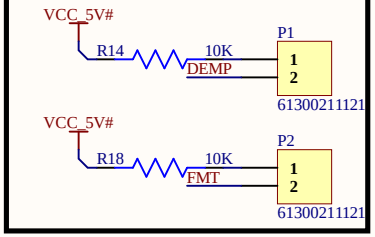
DECOUPLING CAP



24 BIT DAC CIRCUITRY



FMT & DEMP CONTROL

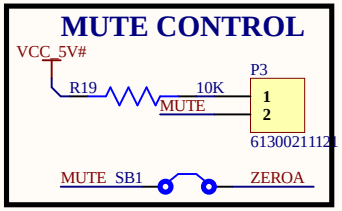


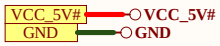
Supporting Circuitry



MUTE OFF - LOW
MUTE ON - HIGH
FMT LOW = 16– to 24–bit, I2S format
FMT HIGH =16–bit right–justified
DEMP HIGH = OFF
DEMP LOW = ON

MUTE CONTROL



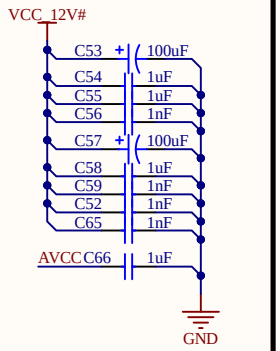


Cannot open file D:\eSpark\eSpark-Hardware\Altium Template\espark-consulting-group-150x150.jpg. File does not exist.	Title: Ezlo_Voice_Assistant_Node.PrjPcb		
	Size: A4	Number: 1	Revision: 1
	Date: 10/20/2022	Sheet 7 of 9	
	File: C:\GitHub\..\DAC_Section.SchDoc	Drawn By: Huzaifa Ali	

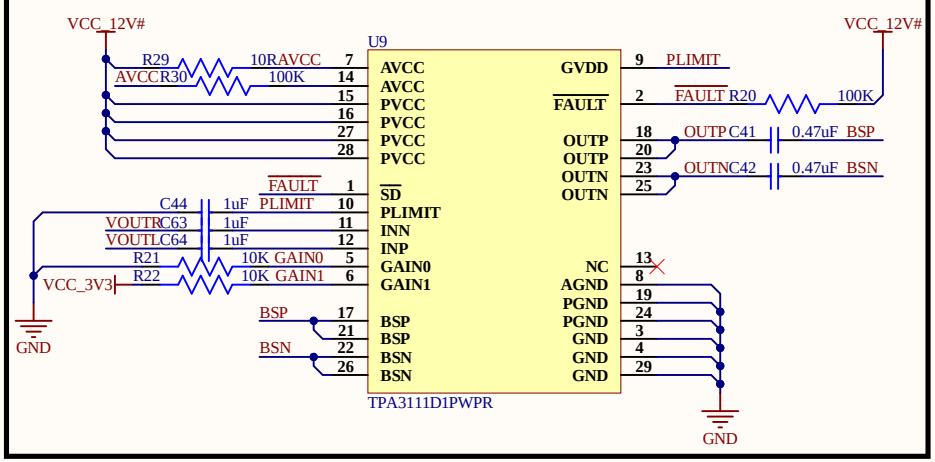
GAIN SIG
GAIN0
GAIN1

DAC_VOUT
VOUTL
VOUTR

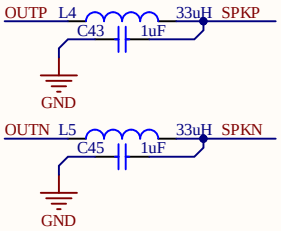
DECOUPLING CAP



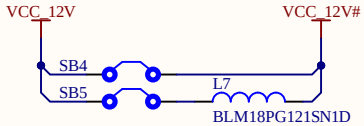
10W 8OHMS AMPLIFIER IC



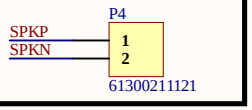
LPF FOR 8OHMs SPEAKER



Power filter



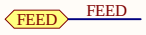
SPEAKER OUTPUT



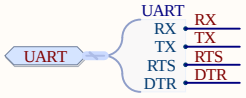
VCC_12V
GND

Cannot open file D:\eSpark\eSpark-Hardware\Altium Template\espark-consulting-group-150x150.jpg. File does not exist.

Title	Ezlo_Voice_Assistant_Node.PrjPcb		
Size	Number		Revision
A4	1		1
Date:	10/20/2022	Sheet 8 of	9
File:	C:\GitHub\..\Amplifier_Section.SchDoc	Drawn By:	Huzaifa Ali



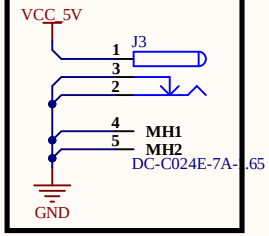
PROGRAM INPUTS



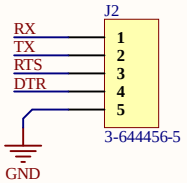
DEBUGGING SIGNALS



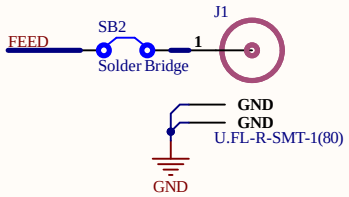
INPUT JACK



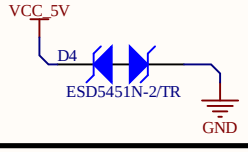
PROGRAMMING HEADER



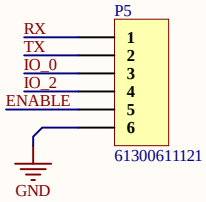
UFL CONECTOR EXTERNAL ANTENNA



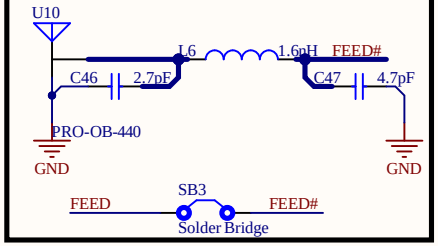
ESD PROTECTION



DEBUG CONNECTOR



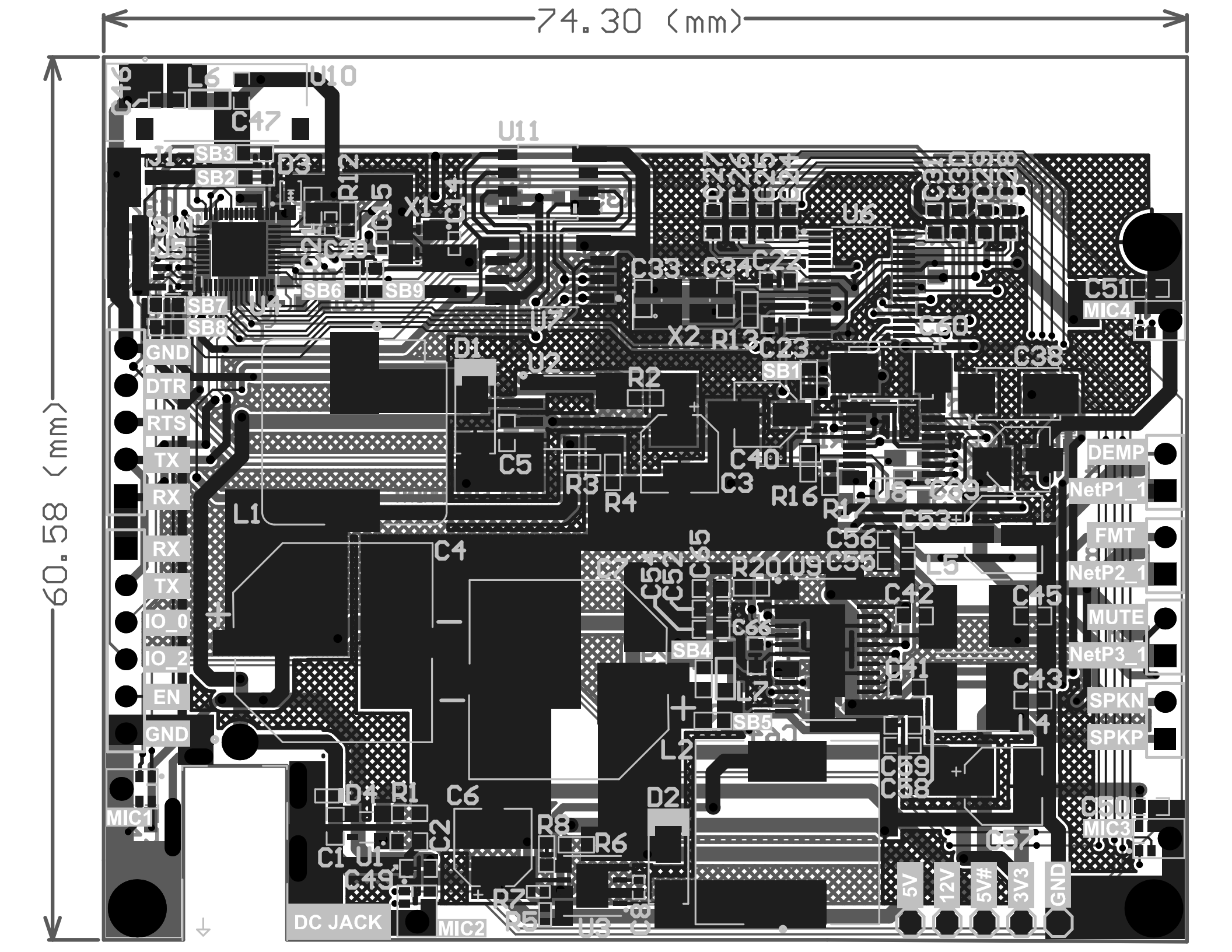
ON-BOARD PCB ANTENNA





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Title	Ezlo_Voice_Assistant_Node.PrjPcb		
Size	Number		Revision
A4	1		1
Date:	10/20/2022	Sheet 9 of	9
File:	C:\GitHub\..\Connector_Section.SchDoc	Drawn By:	Huzaifa Ali


74.30 (mm)
60.58 (mm)
U10
L6
C47
U11
SB3
SB2
D3
R12
C5
X1
C3
C20
SB6
SB9
SB7
SB8
U4
U7
C33
C34
C22
X2
R13
C23
SB1
C51
MIC4
C38
GND
DTR
RTS
TX
RX
RX
TX
IO_0
IO_2
EN
GND
D1
U2
R2
C5
R3
R4
C40
C3
R16
R17
L1
C4
C53
C56
C55
R20109
L5
C42
C45
C66
SB4
C41
C43
L7
SB5
L4
L2
C68
C69
D+
R1
C6
D2
C2
R8
R6
C1
U1
C49
R7
R5
U3
C8
C57
C50
MIC3
MIC1
DC JACK
MIC2
5V
12V
5V#
3V3
GND
DEMP
NetP1_1
FMT
NetP2_1
MUTE
NetP3_1
SPKN
SPKP